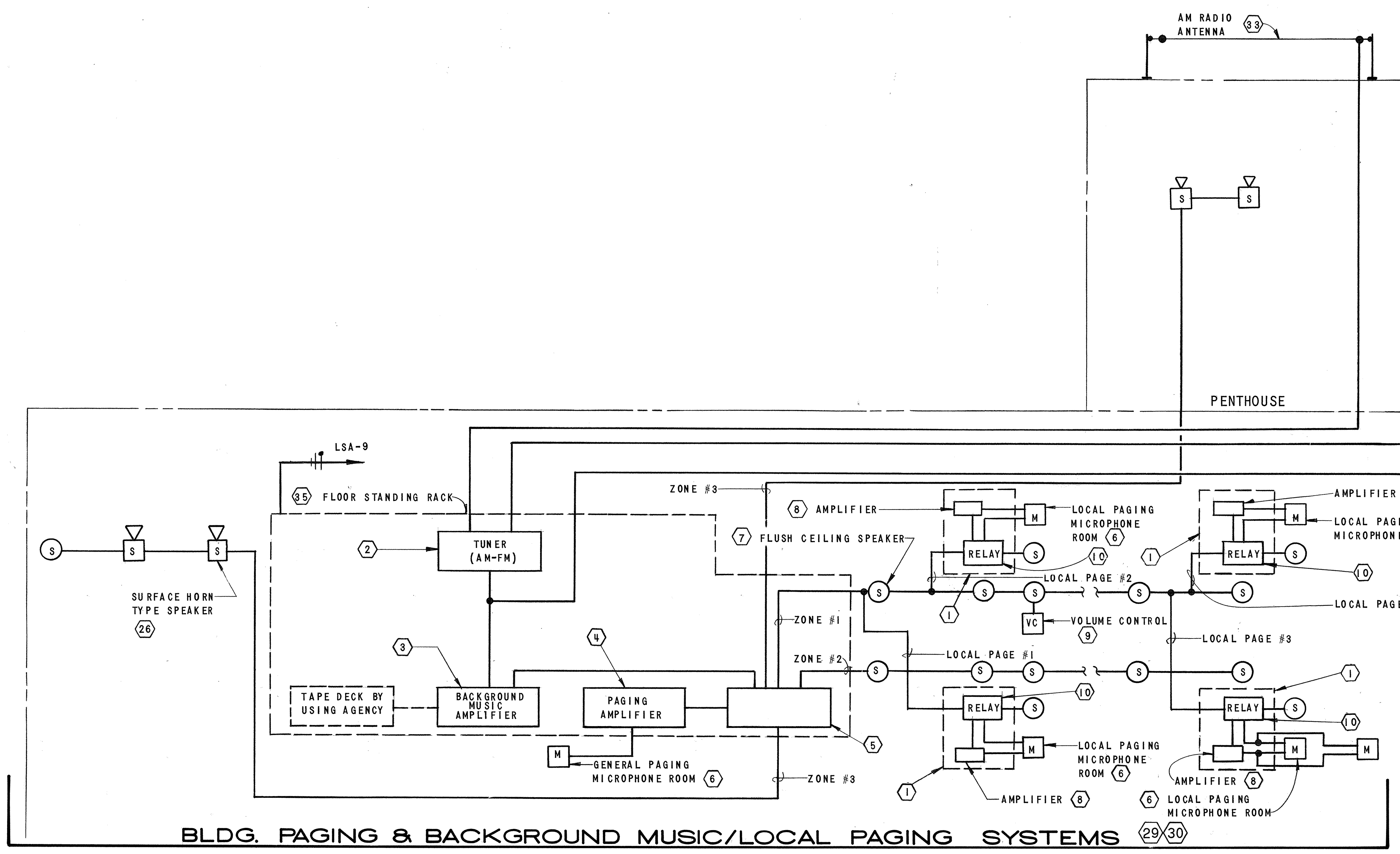
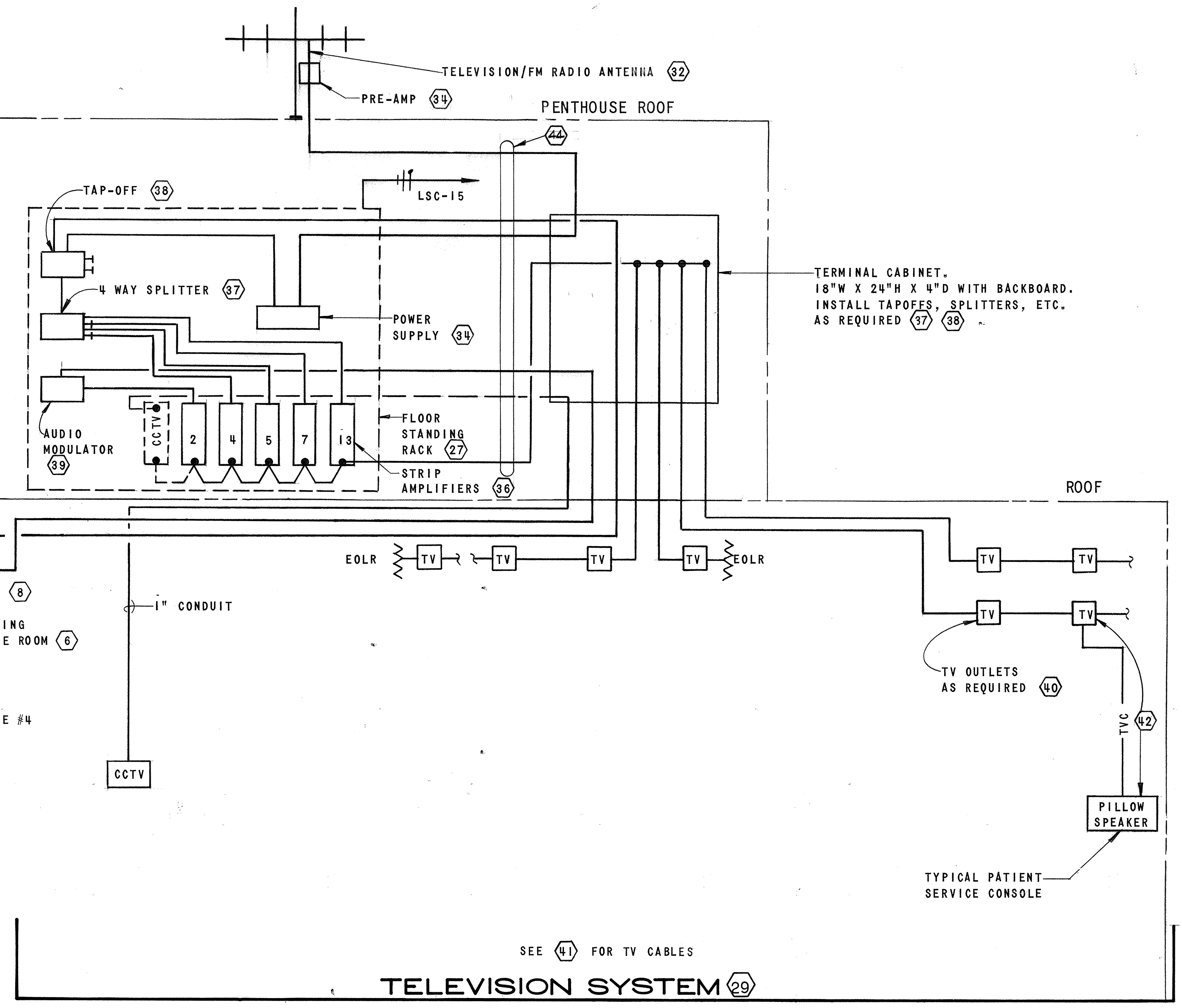
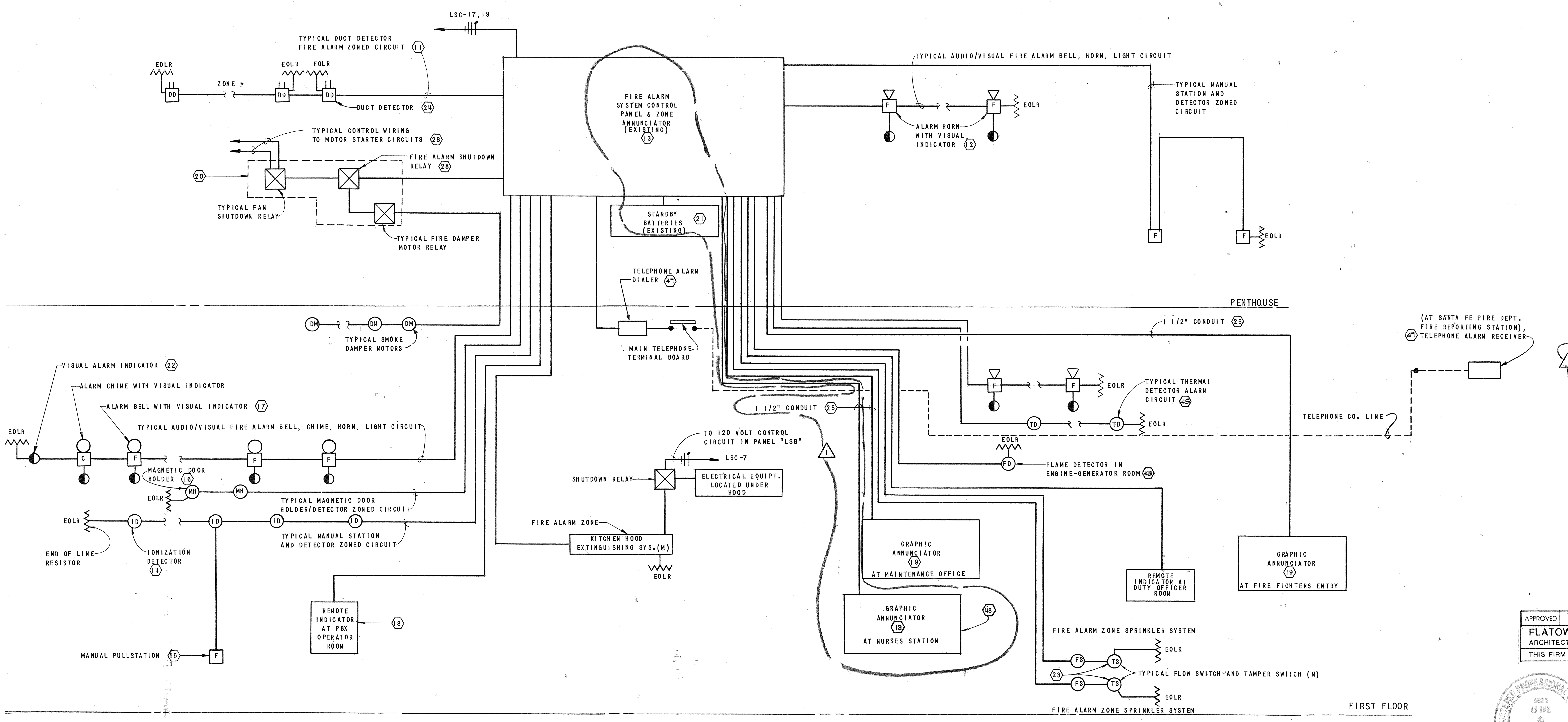

AM RADIO ANTENNA (33)
PENTHOUSE
LSA-9
(35) FLOOR STANDING RACK
ZONE #3
(8) AMPLIFIER
LOCAL PAGING MICROPHONE ROOM (6)
(7) FLUSH CEILING SPEAKER
AMPLIFIER (8)
LOCAL PAGING MICROPHONE ROOM (6)
TUNER (AM-FM)
(2)
RELAY
(10)
LOCAL PAGE #2
LOCAL PAGE #4
SURFACE HORN TYPE SPEAKER (26)
ZONE #1
VOLUME CONTROL (9)
VC
LOCAL PAGE #3
(3)
(4)
ZONE #2
LOCAL PAGE #1
TAPE DECK BY USING AGENCY
BACKGROUND MUSIC AMPLIFIER
PAGING AMPLIFIER
(5)
GENERAL PAGING MICROPHONE ROOM (6)
ZONE #3
AMPLIFIER (8)
(6) LOCAL PAGING MICROPHONE ROOM
S
M
(1)

BLDG. PAGING & BACKGROUND MUSIC/LOCAL PAGING SYSTEMS (29)(30)


TELEVISION/FM RADIO ANTENNA (32)
PRE-AMP (34)
PENTHOUSE ROOF
(44)
TAP-OFF (38)
LSC-15
4 WAY SPLITTER (37)
POWER SUPPLY (34)
TERMINAL CABINET.
18"W X 24"H X 4"D WITH BACKBOARD.
INSTALL TAPOFFS, SPLITTERS, ETC.
AS REQUIRED (37) (38)
AUDIO MODULATOR (39)
CCTV
2
4
5
7
13
FLOOR STANDING RACK (27)
STRIP AMPLIFIERS (36)
ROOF
EOLR
TV
1" CONDUIT
TV OUTLETS AS REQUIRED (40)
TVC (42)
CCTV
PILLOW SPEAKER
TYPICAL PATIENT SERVICE CONSOLE
SEE (41) FOR TV CABLES

TELEVISION SYSTEM (29)


LSC-17,19
TYPICAL DUCT DETECTOR FIRE ALARM ZONED CIRCUIT (11)
EOLR
ZONE #
DD
DUCT DETECTOR (24)
FIRE ALARM SYSTEM CONTROL PANEL & ZONE ANNUNCIATOR (EXISTING) (13)
TYPICAL AUDIO/VISUAL FIRE ALARM BELL, HORN, LIGHT CIRCUIT
TYPICAL MANUAL STATION AND DETECTOR ZONED CIRCUIT
TYPICAL CONTROL WIRING TO MOTOR STARTER CIRCUITS (28)
FIRE ALARM SHUTDOWN RELAY (28)
(20)
TYPICAL FAN SHUTDOWN RELAY
TYPICAL FIRE DAMPER MOTOR RELAY
ALARM HORN WITH VISUAL INDICATOR (12)
F
EOLR
STANDBY BATTERIES (EXISTING) (21)
TELEPHONE ALARM DIALER (47)
PENTHOUSE
DM
TYPICAL SMOKE DAMPER MOTORS
MAIN TELEPHONE TERMINAL BOARD
1 1/2" CONDUIT (25)
(AT SANTA FE FIRE DEPT. FIRE REPORTING STATION), (47) TELEPHONE ALARM RECEIVER
VISUAL ALARM INDICATOR (22)
ALARM CHIME WITH VISUAL INDICATOR
ALARM BELL WITH VISUAL INDICATOR (17)
TYPICAL AUDIO/VISUAL FIRE ALARM BELL, CHIME, HORN, LIGHT CIRCUIT
C
1 1/2" CONDUIT (25)
TYPICAL THERMAL DETECTOR ALARM CIRCUIT (45)
TD
TELEPHONE CO. LINE
TO 120 VOLT CONTROL CIRCUIT IN PANEL "LSB"
LSC-7
MAGNETIC DOOR HOLDER (16)
MH
SHUTDOWN RELAY
ELECTRICAL EQUIPT. LOCATED UNDER HOOD
FLAME DETECTOR IN ENGINE-GENERATOR ROOM (46)
FD
TYPICAL MAGNETIC DOOR HOLDER/DETECTOR ZONED CIRCUIT
ID
TYPICAL MANUAL STATION AND DETECTOR ZONED CIRCUIT
END OF LINE RESISTOR
IONIZATION DETECTOR (14)
FIRE ALARM ZONE
KITCHEN HOOD EXTINGUISHING SYS. (M)
GRAPHIC ANNUNCIATOR (19) AT MAINTENANCE OFFICE
REMOTE INDICATOR AT DUTY OFFICER ROOM
GRAPHIC ANNUNCIATOR (19) AT FIRE FIGHTERS ENTRY
REMOTE INDICATOR AT PBX OPERATOR ROOM
(18)
GRAPHIC ANNUNCIATOR (19) AT NURSES STATION
(48)
FIRE ALARM ZONE SPRINKLER SYSTEM
FS
TS
(23)
TYPICAL FLOW SWITCH AND TAMPER SWITCH (M)
MANUAL PULLSTATION (15)
FIRST FLOOR
1

FIRE ALARM SYSTEM RISER DIAGRAM (29)(31)

NO SCALE

NOTES

1. INSTALL EQUIPMENT IN COMMON SYSTEM RACK (20).
2. BOGEN #TP160 WITH #RPK-31 RACK KIT.
3. ALTEC #1608A-150W WITH 2#15095 BRIDGING TRANSFORMER.
4. ALTEC #1608A-150W WITH #1588C PREAMP.
5. SOUNDOLIER #MPC-206.
6. BOGEN #SHURE 450.
7. SOUNDOLIER #C10T25 SPEAKER WITH #T720-8 AND #T95-8 BACKBOX AND BAFFLE.
8. BOGEN #CHS-20A WITH TM200 INPUT TRANSFORMER.
9. SOUNDOLIER #AT-10.
10. SOUNDOLIER #RP22-4 RELAY WITH #PS24-07 POWER SUPPLY.
11. JOHNSON CONTROLS INC. CHIME #SAU-1-2, 24 VOLT POLARIZED/SUPERVISED; #SAU-1-3 WITH E.O.L.R.; VISUAL INDICATOR #SFA-12 SERIES.
12. JOHNSON CONTROLS INC. HORN (97db) #SAU-4-5, 24 VOLT D.C. POLARIZED/SUPERVISED; #SAU-4-6 WITH E.O.L.R.; VISUAL INDICATOR #SFA-12 SERIES.
13. FIRE ALARM CONTROL PANEL-JOHNSON CONTROLS INC. MODEL #SET-7000 SERIES. REFER TO FLOOR PLANS FOR FIRE ALARM ZONE REQUIREMENTS. △1
14. JOHNSON CONTROLS INC. IONIZATION DETECTOR #SMU-20-4, FLUSH MOUNTED.
15. JOHNSON CONTROLS INC. MANUAL FIRE ALARM STATION #SFA-10-2.
16. MAGNETIC DOOR HOLDER WITH BUILT IN SMOKE DETECTOR IS BEING FURNISHED UNDER ARCHITECTURAL SECTION OF SPECIFICATIONS. COORDINATE VOLTAGE OF HOLDER AND INCORPORATE POWER, ALARM, AND DOOR RELEASE CONNECTIONS INTO FIRE ALARM SYSTEM.
17. JOHNSON CONTROLS INC. BELL #SAU-3-1, 24 VOLTS D.C. POLARIZED/SUPERVISED; VISUAL INDICATOR #SFA-12 SERIES.
18. JOHNSON CONTROLS INC. TROUBLE INDICATOR PANEL #SFA-9-7, FLUSH MOUNTED, COMPLETE WITH PILOT LIGHT, SILENCING SWITCH, RESET SWITCH, ALARM TEST SWITCH AND LAMP SUPERVISION.
19. JOHNSON CONTROLS INC. REMOTE GRAPHIC ANNUNCIATOR #SAN-402 WITH SBX-1-6 FLUSH OUTLET BOX. 60 LAMP UNIT WITH LAMP SUPERVISION. △1
20. INSTALL RELAYS IN COMMON ENCLOSURE ADJACENT TO FIRE ALARM CONTROL PANEL.
21. JOHNSON CONTROLS INC. STAND-BY BATTERY BANK #SBX-1-20 WITH NI-CAD BATTERIES-16AH AT 10 HOUR RATE #SFA-14-3.
22. JOHNSON CONTROLS INC. VISUAL INDICATOR, CEILING MOUNTED #SFA-12 SERIES.
23. VERIFY EXACT LOCATION OF FLOW AND TAMPER SWITCHES WITH SPRINKLER SYSTEM SUPPLIER. WIRE INTO SEPARATE FIRE ALARM ZONE AS REQUIRED.
24. MECHANICAL FURNISHED DETECTOR. COORDINATE WIRING, VOLTAGE AND POWER REQUIREMENTS FOR COMPATABLE SYSTEMS. CONNECT DETECTORS TO SEPARATE ZONE.
25. CONDUIT SIZE INDICATED IS APPROXIMATE. ACTUAL CONDUIT SIZE AND WIRING REQUIREMENTS SHALL BE FURNISHED AND INSTALLED PER SYSTEM MANUFACTURERS REQUIREMENTS. △1
26. ELECTRO-VOICE #PA15AT HORN SPEAKER WITH BUILT-IN TRANSFORMER.
27. SOUNDOLIER #200-42 RACK WITH BUILT-IN FILLER PANELS, LOCKING DOOR AND ACR-193 POWER PANEL AND ROLLER TRUCK #CT1020.
28. REFER TO SHEETS E-19, E-20, & E-21 FOR CONTROL LADDER DIAGRAMS AND INTERCONNECTION REQUIREMENTS
29. CONDUIT AND WIRING SIZES, QUANTITIES, ETC. SHALL BE IN ACCORDANCE WITH APPROVED SYSTEM MANUFACTURERS REQUIREMENT. COORDINATE REQUIREMENTS PRIOR TO BIDS.
30. PAGING SYSTEM SHALL BE DIVIDED INTO THREE MAIN BUILDING ZONES ALONG WITH GENERAL PAGE AND FOUR LOCAL ZONES. IN GENERAL, ZONING IS INDICATED ON FLOOR PALNS. THE PATIENT CARE AREA SHALL BE ONE ZONE, ADMINISTRATIVE AREA A SECOND ZONE AND BUILDING MAINTENANCE & EQUIPMENT AREAS THE THIRD ZONE. LOCAL PAGING WILL BE IN LOCATIONS INDICATED ON DRAWINGS.
31. FIRE ALARM ZONES ARE INDICATED ON PLANS. BELLS, CHIMES, HORNS, ETC. WILL BE CONNECTED FOR GENERAL ALARM THROUGHOUT THE FACILITY.
32. JERROLD #J-283X-TV AND FM ANTENNA WITH MULTIPOLE GUYS ON 20'-0" RIGID 1 1/2" CONDUIT MAST.
33. TWO 5'-0" POSTS 50'-0" APART INTER CONNECTED WITH AM ANTENNA WIRE LOCATED NEAR EDGE OF ROOF. INSTALL GUY WIRES.
34. JERROLD #42075 POWER SUPPLY AND PRE-AMP.
35. SOUNDOLIER #200-61 RACK WITH BLANK FILLER PANELS, LOCKING DOOR AND ACR-193 POWER PANEL.
36. JERROLD "HPM" SERIES STRIP AMPLIFIERS.
37. SPLITTERS SHALL BE EQUAL TO JERROLD #1596B (TWO WAY) AND #1597 (FOUR WAY).
38. TAP-OFFS SHALL BE EQUAL TO JERROLD #1407A.
39. JERROLD #AT S-2 AUDIO MODULATOR.
40. TV OUTLETS SHALL BE PROVIDED WITH COMPATABLE CONNECTORS. SINGLE SET FEEDS SHALL BE JERROLD #UT-12, MULTIPLE SET FEEDS JERROLD #UT-82.
41. TV CABLES SHALL BE JERROLD TYPE "CAC-6".
42. REMOTE TV CONTROL IN PATIENT SERVICE CONSOLE SHALL BE CONNECTED TO JERROLD #UTS-R OR COMPARABLE TYPE COMPATABLE WITH NURSE CALL/PILLOW SPEAKER SYSTEM BEING INSTALLED.
43. TV CABINETS SHALL BE SURFACE TELEPHONE TERMINAL TYPE 12"W X 18"H X 4"D WITH PLYWOOD BACKBOARD.
44. ROUTE TV SYSTEM CABLES IN COMMON FLEXIBLE CONDUIT BETWEEN TV TERMINAL CABINET AND TV EQUIPMENT RACK.
45. THERMAL DETECTOR, JOHNSON CONTROLS INC. CONBINATION RATE-OF-RISE & FIXED TEMPERATURE #SMU-7 SERIES.
46. FLAME DETECTOR, JOHNSON CONTROLS INC. #SMU-20-17, SURFACE.
47. TELEPHONE DIALER, JOHNSON CONTROLS INC. #KG-7001, TWO CHANNEL; TELEPHONE RECEIVER JOHNSON CONTROLS INC. #KG-7000 SERIES.
48. CONTRACTOR SHALL FURNISH AND INSTALL NEW GRAPHIC ANNUNCIATOR TO MATCH EXISTING. CONTRACTOR SHALL FURNISH ALL REQUIRED RELAYS, LAMPS, RESISTORS, DIODES AND ANY OTHER EQUIPMENT REQUIRED TO ALLOW A COMPLETE FUNCTIONAL SYSTEM. CONTRACTOR SHALL FURNISH ADDITIONAL BATTERIES AS REQUIRED TO COMPENSATE FOR ADDITIONAL CURRENT DRAIN. △1

REV.	DATE	ACTION	DESCRIPTION	BY	APP'D
△1	8-15-78		ADD F.A. ANNUNCIATOR	AEI	REM
	8-4-78		AS-BUILT		

APPROVED	REGIONAL ENGINEER	ARCHITECTURAL	ELECTRICAL	MECHANICAL	STRUCTURAL
		CEL			

APPROVED	STRUCTURAL	ARCHITECTURAL	MECHANICAL	ELECTRICAL

FLATOW MOORE BRYAN & ASSOCIATES
ARCHITECTS ENGINEERS ALBUQUERQUE NEW MEXICO
THIS FIRM IS RESPONSIBLE FOR REVISION(S) NO. 1

DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
OFFICE OF FACILITIES ENGINEERING AND PROPERTY MANAGEMENT
REGIONAL OFFICE OF FACILITIES ENGINEERING AND CONSTRUCTION

IHS/OFEPM MSHFP
INDIAN HEALTH SERVICE FACILITY
SANTA FE COMPREHENSIVE HEALTH FACILITY

R06-16-

DRAWN BY U & L CHECKED BY U & L

W. C. KRUGER and ASSOCIATES
ARCHITECTS • PLANNERS, INC.
ALBUQUERQUE, NEW MEXICO

ELECTRICAL
F. A. AND SPECIAL
SYSTEMS DIAGRAMS

SCALE: NONE DATE: 11/2/76

E-13.1 SHEET OF

UHL & LOPEZ

W. C. KRUGER NO. 47

W. C. KRUGER & ASSOCIATES INC. NO. 909

HOLMAN'S INC. 6-76